

API_SPEC.md

SynChain API Specification

Base URL

/api/v1

Health Check

GET /health

Purpose: Verify backend status.

Response:

```
{ "status": "healthy" }
```

Create Simulation

POST /simulate

Request:

```
{ "product_name": "iPhone", "current_stock": 10000, "demand": 8000, "supplier_delay": 5, "warehouse":  
  "Hyderabad" }
```

Response:

```
{ "simulation_id": "123", "status": "created" }
```

Run Agent Pipeline

POST /simulate/{id}/run

Purpose: Execute all agents.

Response:

```
{ "status": "completed" }
```

Get Results

GET /simulate/{id}/results

Response:

```
{ "forecast": 12000, "recommended_inventory": 14000, "risk": "Medium", "warehouse": "W2" }
```

Get Agent Outputs

GET /simulate/{id}/agents

Response:

```
{ "demand_agent": {}, "inventory_agent": {}, "logistics_agent": {}, "risk_agent": {} }
```

Scenario Simulation

POST /simulate/{id}/scenario

Request:

```
{ "demand_increase": 50 }
```

Response:

```
{ "new_forecast": 18000, "inventory_required": 20000 }
```

External Intelligence

GET /intelligence/news

Purpose: Retrieve latest supply-chain related signals.

Response:

```
[ { "title": "Chip shortage reported", "impact_score": 85 } ]
```